SKIN-NET KNOWLEDGE BASE

Clinical Reference for AI-Assisted Dermatology Detection

Capstone Project — WhatsApp Healthcare Bot | Skin-Net (EfficientNetB3)

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Five Disease Classes Covered in This Document:

- **DISEASE 1: ACNE VULGARIS**
- **DISEASE 2: MELANOMA**
- **DISEASE 3: SCALY LESIONS**
- **DISEASE 4: VITILIGO & PIGMENTATION DISORDERS**
- **DISEASE 5: WARTS (VIRAL LESIONS)**

DISEASE 1: ACNE VULGARIS

OVERVIEW

Acne vulgaris is one of the most prevalent dermatological conditions globally, affecting approximately 85% of people between the ages of 12 and 24. It is a chronic inflammatory skin disease involving the pilosebaceous unit (the hair follicle and its associated sebaceous gland). Acne arises when hair follicles become clogged with sebum (skin oil) and dead skin cells, creating an environment where bacteria can thrive and trigger inflammation.

QUICK REFERENCE

Type	Inflammatory skin disease (Pilosebaceous unit)
Prevalence	~85% of adolescents; affects all age groups
Primary Cause	Clogged follicles + Cutibacterium acnes bacteria
Common Sites	Face, neck, chest, back, shoulders
AI Detection	CNN-based grading of comedones, papules, pustules, nodules

PATHOPHYSIOLOGY

The development of acne involves four key mechanisms: (1) Increased sebum production driven by androgens, (2) Abnormal follicular keratinization causing blockage, (3) Proliferation of Cutibacterium acnes (formerly Propionibacterium acnes) bacteria, and (4) Inflammatory response triggered by bacterial byproducts. Hormonal changes — particularly during puberty, menstruation, or stress — stimulate sebaceous glands to overproduce oil, setting off this cascade.

CLASSIFICATION & LESION TYPES

- Comedones (non-inflammatory): Whiteheads (closed) and blackheads (open) — basic blocked follicles
- Papules: Small, raised, red bumps — early inflammatory lesions
- Pustules: Papules filled with pus — active bacterial infection
- Nodules: Large, solid, painful lumps deep in the skin — severe form
- Cysts: Pus-filled lumps — most severe; likely to scar
- Severity grading: Mild (few non-inflammatory), Moderate (mixed), Severe (nodular/cystic)

TREATMENT APPROACHES

- Topical retinoids (tretinoin, adapalene) — normalize follicular keratinization
- Benzoyl peroxide — kills C. acnes bacteria, reduces inflammation
- Topical/oral antibiotics — target bacterial overgrowth (clindamycin, doxycycline)
- Hormonal therapy — oral contraceptives or spironolactone for hormonal acne
- Isotretinoin (Accutane) — powerful oral retinoid for severe, resistant acne

RELEVANCE TO SKIN-NET (AI MODEL)

In the context of your Skin-Net EfficientNetB3 model, acne classification typically focuses on identifying the distribution and type of lesions. Key challenges include distinguishing between pustular acne and

early stage folliculitis, and grading severity from a single image. ABCD rule features are less relevant here; instead, texture and distribution patterns dominate. The PAD-UFES-20 and DermNet datasets contain clinical acne images suitable for training.

DATASETS & RESOURCES FOR KNOWLEDGE BASE ENRICHMENT

- DermNet Acne Dataset (Kaggle): https://www.kaggle.com/datasets/shubhamgoel27/dermnet
- ACNE04 Dataset (Graded acne images): https://github.com/xpwu95/LDL
- ISIC Archive (broad skin conditions): https://www.isic-archive.com/
- DermNet NZ Clinical Reference: https://dermnetnz.org/topics/acne

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DISEASE 2: MELANOMA

OVERVIEW

Melanoma is the most dangerous form of skin cancer, originating in melanocytes — the pigment-producing cells of the skin. Although it accounts for only about 1% of all skin cancers, melanoma is responsible for the majority of skin cancer deaths. It is highly aggressive and, if detected late, spreads rapidly to lymph nodes and other organs. Early detection, ideally by dermoscopic AI analysis, can raise the 5-year survival rate from around 25% (late stage) to over 98% (stage I).

QUICK REFERENCE

Type	Malignant cancer of melanocytes
Prevalence	~325,000 new cases/year globally
Risk Factors	UV exposure, fair skin, family history, many moles
Survival Rate	Stage I: ~98% | Stage IV: ~15-25%
AI Detection	CNN + ABCD rule features, dermoscopy classification

PATHOPHYSIOLOGY

Melanoma arises from mutations in melanocytes, typically triggered by ultraviolet (UV) radiation from the sun or tanning beds. Key driver mutations include BRAF (V600E) — present in ~50% of melanomas — and NRAS and NF1 mutations. These mutations disrupt normal cell cycle regulation, causing uncontrolled proliferation. Melanoma can develop in existing moles or appear as new pigmented lesions. It has a high tendency for lymphatic and hematogenous spread.

ABCDE CLINICAL CRITERIA

- A — Asymmetry: One half does not match the other
- B — Border: Irregular, ragged, notched, or blurred edges
- C — Color: Varied shades of brown, black, red, white, or blue
- D — Diameter: Larger than 6mm (about the size of a pencil eraser)
- E — Evolving: Any change in size, shape, color, or new symptoms

STAGING (TNM SYSTEM)

- Stage 0 (Melanoma in situ): Confined to epidermis — nearly 100% curable
- Stage I: Thin tumors (<2mm) — high cure rate with surgical excision
- Stage II: Thicker tumors, possible ulceration — higher recurrence risk
- Stage III: Spread to nearby lymph nodes — immunotherapy/targeted therapy
- Stage IV: Distant metastasis to lungs, liver, brain — systemic treatment required

TREATMENT APPROACHES

- Surgical excision with wide margins — primary treatment for early stages
- Sentinel lymph node biopsy — checks for regional spread
- Targeted therapy: BRAF/MEK inhibitors (vemurafenib, dabrafenib) for BRAF-mutant melanoma

- Immunotherapy: PD-1 inhibitors (pembrolizumab, nivolumab) — improve long-term survival
- Radiation therapy — used for brain metastases or adjuvant control

RELEVANCE TO SKIN-NET (AI MODEL)

Melanoma is the most studied condition in dermatology AI. Your EfficientNetB3 model must distinguish melanoma from benign pigmented lesions such as seborrheic keratosis or dysplastic nevi. Key feature patterns include irregular pigment networks, atypical vascular structures, and regression zones visible in dermoscopy. The ISIC challenge datasets are the gold standard for this class. Grad-CAM on your model should highlight pigment network disruptions.

DATASETS & RESOURCES FOR KNOWLEDGE BASE ENRICHMENT

- ISIC 2020 Challenge Dataset (33K images): https://www.kaggle.com/competitions/siim-isic-melanoma-classification
- HAM10000 Dataset (10K dermoscopy images): https://www.kaggle.com/datasets/kmader/skin-lesion-analysis-toward-melanoma-detection
- MIDAS Multimodal Dataset (NEJM AI): https://ai.nejm.org/doi/full/10.1056/AIdbp2400732
- PH2 Dermoscopy Dataset: https://www.fc.up.pt/addi/ph2%20database.html
- ISIC Archive (Official): https://www.isic-archive.com/

DISEASE 3: SCALY LESIONS

OVERVIEW

Scaly lesions encompass a group of chronic inflammatory skin diseases characterized by abnormal skin cell turnover leading to scaling, thickening, and flaking of the skin. The major conditions under this umbrella include Psoriasis (immune-mediated plaques), Eczema/Atopic Dermatitis (allergy-driven inflammation), Seborrheic Dermatitis (yeast-driven scalp/face scaling), and Tinea Versicolor (fungal). In your Skin-Net model, this class captures the shared visual feature of epidermal scaling regardless of the underlying mechanism.

QUICK REFERENCE

Type	Chronic inflammatory / hyperproliferative skin diseases
Prevalence	Psoriasis ~2-3% globally; Eczema ~15-20% of children
Primary Cause	Immune dysregulation, genetic predisposition, triggers
Common Sites	Scalp, elbows, knees, face (varies by condition)
AI Detection	Plaque texture, scale pattern, distribution features

PSORIASIS — DETAILED OVERVIEW

Psoriasis is an autoimmune condition where T-cell activation causes keratinocytes to proliferate 6-10x faster than normal (turnover in 3-5 days vs. 28-30 days). This results in the classic silvery-white plaques on erythematous (red) bases. The IL-17 and IL-23 cytokine pathways are central to its pathogenesis. Psoriasis is lifelong and episodic — flares can be triggered by stress, infections, medications (beta-blockers, lithium), or alcohol.

ECZEMA (ATOPIC DERMATITIS) — DETAILED OVERVIEW

Atopic dermatitis results from a defective skin barrier (often due to filaggrin gene mutations) combined with a Th2-skewed immune response. This allows allergens and irritants to penetrate, triggering an IgE-mediated and cytokine-driven inflammatory cascade. The hallmark is intensely pruritic (itchy), dry, weeping skin. It commonly appears in skin folds (antecubital, popliteal). It is part of the 'atopic triad' with asthma and allergic rhinitis.

KEY DISTINGUISHING VISUAL FEATURES

- Psoriasis: Well-defined silvery-white scales on erythematous plaques; predilection for extensor surfaces
- Eczema: Ill-defined, weeping, crusted lesions; flexural distribution; xerosis (dry skin)
- Seborrheic Dermatitis: Greasy yellowish scales on oily areas — scalp, T-zone of face, chest
- Tinea Versicolor: Hypo/hyperpigmented scaly patches on trunk — fungal origin
- Scale color is a key ML feature: silvery vs. yellowish vs. white/flaky

TREATMENT APPROACHES

- Topical corticosteroids — mainstay for psoriasis and eczema flares

- Vitamin D analogues (calcipotriol) — slow keratinocyte proliferation in psoriasis
- Calcineurin inhibitors (tacrolimus) — non-steroidal option for sensitive areas
- Biologics (IL-17, IL-23, TNF-alpha inhibitors) — for moderate-severe psoriasis
- Dupilumab — IL-4/IL-13 inhibitor approved for moderate-severe atopic dermatitis
- Antifungal shampoos/creams (ketoconazole) — for seborrheic dermatitis

RELEVANCE TO SKIN-NET (AI MODEL)

Scaly lesions are visually overlapping, making this one of the hardest classes in your model. Your EfficientNetB3 will need to learn texture gradients (scale thickness, distribution), color (erythema intensity), and lesion boundary characteristics to differentiate sub-types. Augmentation with varied lighting is critical since scale sheen can confuse the model. Consider adding a confidence threshold for this class given inter-class similarity.

DATASETS & RESOURCES FOR KNOWLEDGE BASE ENRICHMENT

- DermNet Kaggle Dataset (includes psoriasis, eczema): https://www.kaggle.com/datasets/shubhamgoel27/dermnet
- HAM10000 (seborrheic keratosis subset): https://www.kaggle.com/datasets/kmader/skin-lesion-analysis-toward-melanoma-detection
- Diverse Dermatology Images (Stanford DDI): https://ddi-dataset.github.io/
- SCIN Dataset (Google Research, 10K+ images): https://github.com/google-research-datasets/scin
- DermNet NZ — Psoriasis Reference: https://dermnetnz.org/topics/psoriasis

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DISEASE 4: VITILIGO & PIGMENTATION DISORDERS

OVERVIEW

Vitiligo is a chronic autoimmune skin disorder characterized by progressive depigmentation — the loss of melanocytes in specific areas of skin, resulting in white patches. It affects approximately 0.5-2% of the global population regardless of ethnicity, though it is more visually prominent and psychologically impactful in individuals with darker skin tones. Your folder name 'Vitiligo_Pigmentation' suggests this class may also encompass hyperpigmentation and post-inflammatory pigmentation changes, which are important in the Indian demographic context.

QUICK REFERENCE

Type	Autoimmune depigmentation disorder
Prevalence	~0.5-2% globally; higher in India (~8.8 million cases)
Primary Cause	Autoimmune destruction of melanocytes
Common Sites	Hands, face, genitals, skin folds, areas of friction
AI Detection	Segmentation of depigmented patches; contrast with surrounding skin

PATHOPHYSIOLOGY

The leading theory is autoimmune — CD8+ T-cells, autoreactive to melanocyte antigens like MART-1 and gp100, attack and destroy melanocytes. This is supported by the strong association with other autoimmune conditions (thyroid disease, type 1 diabetes, alopecia areata). Genetic predisposition involves variants in HLA genes and immune regulation genes (PTPN22, FOXP1, NLRP1). Oxidative stress and intrinsic melanocyte fragility may serve as initiating factors before the autoimmune cascade begins.

CLASSIFICATION OF VITILIGO

- Non-segmental (generalized): Most common — bilateral, symmetric patches; progressive course
- Segmental: Unilateral, dermatomal distribution; early onset, stable after initial spread
- Universal: >80% body surface depigmented — rare, severe form
- Focal: Small isolated patches; may represent early generalized vitiligo
- Acrofacial: Affects face and extremities — common initial presentation

PIGMENTATION DISORDERS — RELATED CONDITIONS

- Melasma: Hyperpigmented patches (brown/grey-brown) on face — hormonal/sun-induced
- Post-inflammatory hyperpigmentation (PIH): Dark marks after acne, eczema, or injury
- Albinism: Congenital absence of melanin production (OCA1-4 gene mutations)
- Tinea Versicolor: Yeast-induced peri-follicular hypo/hyperpigmentation on trunk
- Idiopathic guttate hypomelanosis: Small white spots; sun-damaged skin, aging

TREATMENT APPROACHES

- Topical calcineurin inhibitors (tacrolimus, pimecrolimus) — stop immune attack

- Topical/systemic corticosteroids — stabilize disease in active phase
- Narrowband UVB (NB-UVB) phototherapy — stimulate remaining melanocyte stem cells
- Excimer laser (308nm) — targeted phototherapy for stable, focal disease
- Surgical repigmentation (punch grafting, split-thickness grafts) — for stable vitiligo
- JAK inhibitors (ruxolitinib cream — FDA approved 2022) — newest targeted therapy

RELEVANCE TO SKIN-NET (AI MODEL)

Vitiligo presents unique challenges for AI because the diagnostic feature is the absence of pigment, not the presence of a lesion. The model must learn high-contrast white-to-normal-skin boundaries. For Indian skin tones (Fitzpatrick IV-V), the contrast is high and detection is generally easier visually. However, distinguishing vitiligo from pityriasis alba, post-inflammatory hypopigmentation, or tinea versicolor requires learning context. Wood's lamp fluorescence (UV) dramatically enhances vitiligo visibility but won't be available in WhatsApp phone images — this is a real-world limitation to document in your capstone.

DATASETS & RESOURCES FOR KNOWLEDGE BASE ENRICHMENT

- PAD-UFES-20 Dataset (your existing dataset): https://data.mendeley.com/datasets/zr7vgbcyr2/1
- Fitzpatrick 17k (diverse skin tones): https://github.com/mattgroh/fitzpatrick17k
- DDI Dataset — Stanford (biopsy-confirmed): https://ddi-dataset.github.io/
- DermNet NZ — Vitiligo Reference: https://dermnetnz.org/topics/vitiligo
- SCIN — Google Research Dataset: https://github.com/google-research-datasets/scin

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DISEASE 5: WARTS (VIRAL LESIONS)

OVERVIEW

Warts are benign epidermal growths caused by infection with Human Papillomavirus (HPV). They are among the most common dermatological complaints worldwide, particularly in children and immunocompromised individuals. Over 200 HPV types have been identified; different strains cause distinct clinical presentations. Though non-malignant in most cases, certain HPV strains (notably HPV-16 and HPV-18) are known carcinogens — but standard cutaneous warts are caused by low-risk strains like HPV-1, 2, and 4.

QUICK REFERENCE

Type	HPV-induced benign epidermal hyperplasia
Prevalence	~10% of children; up to 20% of school-age kids
Causative Agent	Human Papillomavirus (HPV types 1, 2, 4, 10 etc.)
Transmission	Direct skin contact; auto-inoculation; fomites
AI Detection	Surface texture analysis, hyperkeratotic papule identification

PATHOPHYSIOLOGY

HPV infects keratinocytes in the basal layer through micro-abrasions. The viral E6 and E7 oncoproteins disrupt normal cell cycle regulation by inactivating p53 and pRb tumor suppressors, leading to hyperproliferation of infected keratinocytes. The immune system in healthy individuals eventually clears most warts spontaneously (60-70% clear within 2 years without treatment). The characteristic 'black dots' seen in common warts are thrombosed capillaries within the dermal papillae, displaced by rapid epidermal proliferation.

CLASSIFICATION OF WARTS

- Verruca vulgaris (Common warts): Rough, hyperkeratotic, dome-shaped papules — hands, knees
- Verruca plantaris (Plantar warts): Deep, painful, on soles; may have mosaic pattern
- Verruca plana (Flat warts): Smooth, flat-topped, flesh-colored — face, hands, legs
- Filiform warts: Long, thread-like projections — around eyelids, nose, lips
- Condyloma acuminata (Genital warts): HPV-6 and HPV-11; sexually transmitted
- Epidermodysplasia verruciformis: Rare genetic disorder with widespread HPV infection and malignant risk

CLINICAL MIMICS (DIFFERENTIAL DIAGNOSIS)

- Molluscum contagiosum: Poxvirus-caused; umbilicated papules — different from warts
- Seborrheic keratosis: Stuck-on, waxy appearance — may mimic flat warts in elderly
- Calluses/corns: No black dots; no interrupted skin lines; different location pattern
- Amelanotic melanoma: Can rarely mimic hyperkeratotic wart — biopsy if uncertain
- Squamous cell carcinoma: In immunocompromised patients with chronic 'warts'

TREATMENT APPROACHES

- Watchful waiting — 60-70% of warts resolve spontaneously within 2 years
- Salicylic acid preparations — keratolytic; first-line topical treatment
- Cryotherapy (liquid nitrogen) — freezes and destroys infected keratinocytes
- Immunotherapy (DPCP, imiquimod) — stimulates local immune response against HPV
- Laser ablation (CO2 laser) — for recalcitrant or extensive warts
- HPV vaccine (Gardasil) — prevents high-risk and genital wart HPV strains

RELEVANCE TO SKIN-NET (AI MODEL)

Common warts have distinctive texture features — hyperkeratotic surface, disrupted skin lines, and thrombosed capillary dots — that CNNs can learn well in high-resolution images. However, phone camera images sent via WhatsApp will often lack the resolution to see capillary dots, making texture and shape the primary features. The main challenge is distinguishing warts from calluses and seborrheic keratosis. Plantar warts are particularly hard to classify from phone images given their subsurface growth pattern. Consider this as a real-world accuracy limitation in your capstone discussion.

DATASETS & RESOURCES FOR KNOWLEDGE BASE ENRICHMENT

- PAD-UFES-20 (includes wart-like lesions): https://data.mendeley.com/datasets/zr7vgbcyr2/1
- DermNet Kaggle Dataset: https://www.kaggle.com/datasets/shubhamgoel27/dermnet
- HAM10000 (dermatofibroma, keratosis — visual mimics): https://www.kaggle.com/datasets/kmader/skin-lesion-analysis-toward-melanoma-detection
- DermNet NZ — Warts Reference: https://dermnetnz.org/topics/warts
- ISIC Archive (broader skin lesion datasets): https://www.isic-archive.com/

GENERAL DERMATOLOGY DATASETS & RESOURCES

The following resources are useful across all five disease classes in your Skin-Net model:

MULTI-CLASS DATASETS

- ISIC Archive — International Skin Imaging Collaboration (gold standard): https://www.isic-archive.com/
- DermNet Kaggle (23 skin disease classes, 19.5K images): https://www.kaggle.com/datasets/shubhamgoel27/dermnet
- HAM10000 — 10K dermoscopy images, 7 classes: https://www.kaggle.com/datasets/kmader/skin-lesion-analysis-toward-melanoma-detection
- PAD-UFES-20 — Your primary dataset (Mendeley): https://data.mendeley.com/datasets/zr7vgbcyr2/1
- SCIN by Google Research — 10K+ diverse skin images: https://github.com/google-research-datasets/scin
- Fitzpatrick 17k — Diverse skin tone dataset: https://github.com/mattgroh/fitzpatrick17k
- DDI — Stanford biopsy-confirmed diverse dataset: https://ddi-dataset.github.io/

CLINICAL REFERENCE SITES

- DermNet NZ (Best free clinical dermatology reference): https://dermnetnz.org/
- AAD (American Academy of Dermatology): https://www.aad.org/public/diseases
- VisualDx — Clinical diagnostic decision support: https://www.visualdx.com/
- Skinive — AI skin disease dataset reviews: https://skinive.com/skin-disease-datasets/